

The compound 5,10-methylene tetrahydrofolate (5,10mTHF) is derived from folic acid (vitamin B₉) and is required in the active site of the enzyme thymidylate synthase for thymine nucleotide synthesis. Therefore, 5,10mTHF is most likely:

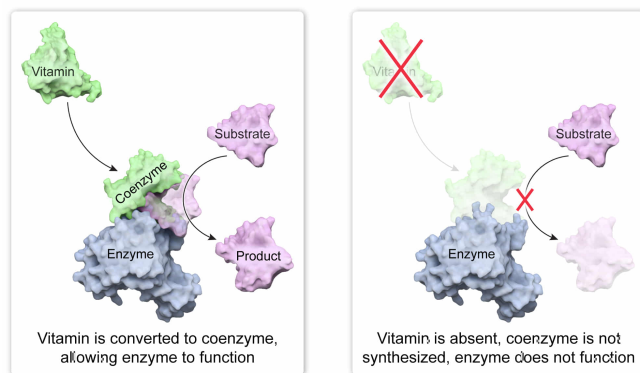
- ☐ A. a transcription factor. [3%]
- ☐ B. an inhibitor.
- ☐ C. an allosteric activator. [10%]
- ✓ ☒ D. a coenzyme. [85%]

Correct

84% answered correctly

Explanation:

Vitamins and cofactors in enzyme function



Enzymes are catalysts that facilitate reactions in biological settings. They consist predominantly of amino acids, but many enzymes **require non-amino acid components** called **cofactors** to function. Many **coenzymes**, which are simply cofactors that are **organic molecules**, are **derived from vital organic compounds** that can be obtained only through the diet, collectively called **vitamins**. Examples of coenzymes include tetrahydrofolate (THF), NADH, FADH₂, and coenzyme A.

The question states that 5,10-methylene tetrahydrofolate (5,10mTHF) is derived from a vitamin (B₉) and that it is required for enzymatic function of thymidylate synthase. Therefore, 5,10mTHF is most likely **acting as a coenzyme**.

(Choice A) **Transcription factors** bind to DNA and alter transcription levels by affecting RNA polymerase binding. They are not involved in the synthesis of individual nucleotides or nitrogenous bases such as thymine.

(Choice B) Inhibitors reduce the activity of an enzyme. According to the question, 5,10mTHF is required for enzymatic function.

(Choice C) Allosteric activators increase the activity of an enzyme by binding somewhere *other than the active site* and changing the shape of the enzyme. The question states that 5,10mTHF is required in the active site, so it does not act allosterically.

Educational objective:

Enzymes frequently require non–amino acid components to carry out their functions. These components are referred to as cofactors and coenzymes, and many of them are derived from vital organic compounds collectively called vitamins.

Time spent:0 sec

Subject:
Biochemistry

QID:402012

System:
Enzymes